

F O R T I M A N A G E R N S E 6

Lab 02 — Administrative Domains

01. Architecture and Compatibility

► 1.1 What FortiManager is made of

FortiManager is three things stacked together: an appliance with its own system settings, a management module holding a copy of every managed device's configuration, and an ADOM layer that partitions that management into independent groups.

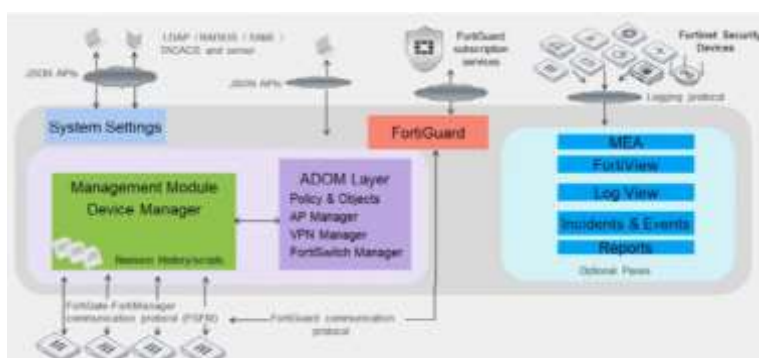


Figure 1 — FortiManager architecture — system settings, the Device Manager module, the ADOM layer, and the optional logging panes.

The lower half of that diagram carries the two protocols that matter. **FGFM** — the FortiGate-to-FortiManager protocol, on TCP 541 — is the management tunnel. **The FortiGuard protocol** is the separate path by which FortiManager obtains signature and rating updates and redistributes them to managed devices.

That second role is worth stating plainly: **FortiManager can act as a local update distribution point, much as a WSUS server does for Windows.** Devices pull their updates from FortiManager rather than reaching FortiGuard individually — which is the only workable model when managed firewalls have no direct internet path, as is normal in an industrial network.

► 1.2 FortiGuard deployment models

Model	How updates reach devices
Open network	Each device reaches FortiGuard directly. Simplest, and requires outbound internet from every firewall.
Cascade	FortiManager obtains updates from FortiGuard and redistributes them to managed devices. The devices themselves need no internet access.
Air-gapped	No path to FortiGuard at all. Update packages are imported manually and distributed from FortiManager.

Cascade is the common enterprise answer and air-gapped the common OT one. Both place FortiManager on the boundary as the only system permitted to talk outward, which is also why its own patch state and access controls matter more than those of the devices behind it.

► 1.3 Version compatibility

A FortiManager does not manage every FortiOS version. Compatibility is published per release pair, and checking it before a deployment is faster than diagnosing a device that registers and then behaves oddly.

Figure 2 — Fortinet's Compatibility Tool — FortiManager 7.6.6 checked against FortiOS 7.0.

FORTIGATE / FORTIOS VERSION	COMPATIBILITY	NOTES
7.0.19	✓ Supported	
7.0.18	✓ Supported	
7.0.17	✓ Supported	
7.0.16	✓ Supported	
7.0.15	✓ Supported	
7.0.14	✓ Supported	
7.0.13	✓ Supported	
7.0.12	✓ Supported	
7.0.11	✓ Supported	
7.0.10	✓ Supported	

Figure 3 — Result: every 7.0 patch release is supported by this FortiManager.

This matters again in section 7, because an ADOM is pinned to a FortiOS version and a device can only join an ADOM that matches its firmware. Compatibility and ADOM version are separate constraints, and both have to be satisfied.

► 1.4 Remote GUI access

FortiManager can proxy a managed FortiGate's own web interface, so an administrator reaches the firewall's GUI through FortiManager rather than connecting to the device directly. On a segmented network where the firewalls are not reachable from the administrator's subnet, this is often the only practical path to a device's native interface.



Figure 4 — Remote GUI access enabled, served on port 8082.

► 1.5 Event Log and Task Monitor

	Records
Event Log	What happened on the appliance — logins, configuration changes, licence validation, system events. An audit trail, written continuously.
Task Monitor	What FortiManager is doing on behalf of an administrator — installs, retrieves, imports, backups. Each task has a lifecycle, a progress state, and a success or failure result.

The practical distinction: when an install appears to do nothing, the Task Monitor says whether it ran and what it returned; the Event Log says whether anything else happened at the same time. A failed task produces a Task Monitor entry with an error string, and that string is usually the whole diagnosis — as section 8 demonstrates.

02. Administrative Domains

An ADOM is a virtual partition inside one FortiManager. Devices, policy packages, objects and administrators all belong to an ADOM, and nothing crosses between them. The useful mental model is a zone: a Cairo ADOM has its own administrators, its own firewalls and its own policy, and an administrator scoped to it cannot see anything else on the appliance.

A FortiGate belongs to exactly one ADOM. An administrator can be granted access to several. That asymmetry is what makes the model work — devices are partitioned, people are not necessarily.

► 2.1 Enabling ADOMs

ADOMs are disabled by default and only a super-admin account can enable them. Enabling ends the current session, because the navigation changes: Device Manager and Policy & Objects gain an ADOM selector, and every subsequent action happens inside a chosen ADOM.



Figure 5 — System Settings → ADOMs, before enabling. Only root exists.

```
config system global
    set adom-status enable
end
```



Figure 6 — After re-login: the ADOM selector, offering root and Global Database.

► 2.2 The default ADOMs

root is the default working ADOM and is typed for FortiGate devices — it manages FortiGates and nothing else. Alongside it, FortiManager pre-creates an ADOM for each supported device family it can manage.



Figure 7 — The full default ADOM list — FortiGate-family ADOMs under Central Management, and separate ADOMs for other device types.

Group	Contents
Central Management	root, FortiProxy, FortiFirewall, FortiFirewallCarrier, FortiCarrier, and Global Database — the families FortiManager manages centrally.
Backup Mode	Empty until an ADOM is created in backup mode. ADOMs are grouped here by their operation mode, not by device type.
Other Device Types	FortiAnalyzer, FortiMail, FortiWeb, FortiSandbox, FortiClient, FortiAuthenticator, Syslog and others — logging and reporting rather than configuration management.

Global Database is not a working ADOM. It holds global policy packages and objects that are assigned down into other ADOMs, and it is the subject of a later lab.

► 2.3 What the licence actually counts

The trial licence on this appliance allows two ADOMs and three devices, which raised a question worth answering before designing anything around it: do the default ADOMs consume the allowance?

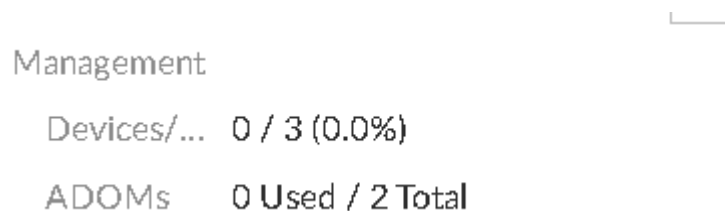


Figure 8 — Before: 0 of 2 ADOMs used, with root and every default ADOM already present.

```

FortiManager-Laps # get system status
Platform Type           : FMG-UM64
Platform Full Name      : FortiManager-UM64
Version                 : v7.6.2-build3415 241212 (GA.F)
Serial Number           : FMG-UMTM26011676
BIOS version            : 04000002
Hostname                : FortiManager-Laps
Max Number of Admin Domains : 2
Max Number of Device Groups : 3
Admin Domain Configuration : Disabled
FIPS Mode                : Disabled
HA Mode                  : Stand Alone
Branch Point             : 3415
Release Version Information : GA.F
Current Time              : Sun Oct 04 05:33:10 GST 2026
Daylight Time Saving      : Yes
Time Zone                 : (GMT+4:00) Abu Dhabi, Muscat.
x86-64 Applications      : Yes
Disk Usage                : Free 470.50GB, Total 491.00GB
File System                : Ext4
License Status             : Valid
Image Signature           : Image is GA Certified

```

Figure 9 — get system status, reporting a maximum of 2 administrative domains and 3 device groups.

Question	Whether root and the pre-created default ADOMs count against the licensed ADOM limit.
Observation	With root, Global Database and sixteen default ADOMs present, the licence widget read 0 Used / 2 Total. After creating one custom ADOM it read 1 Used / 2 Total.
Conclusion	Only administrator-created ADOMs consume the licence. The defaults are templates rather than allocations.
Consequence	This appliance can hold two custom ADOMs alongside root — enough to demonstrate separation, scoping and migration, but not enough for a global ADOM above two child ADOMs.

03. ADOM Operation Mode

Operation mode is chosen when an ADOM is created and governs whether FortiManager may push configuration to the devices inside it.



Figure 10 — The Mode selector — Normal or Backup.

Mode	Behaviour
Normal (default)	Full management. FortiManager holds the device database, administrators edit it, and changes are installed to the device. Configuration changes made on the FortiGate are reported back over FGFM and recorded as a new revision.
Backup	Read-only management. FortiManager collects configuration revisions from the device but cannot install changes to it. Used where a team wants revision history and disaster recovery without surrendering day-to-day control of the firewall.

Backup mode is not a lesser version of normal mode — it answers a different question. An organisation that wants central visibility and a recoverable configuration history, but whose firewall team intends to keep editing devices directly, wants backup mode. Deploying normal mode into that situation produces constant out-of-sync states and a management system nobody trusts.

The mode is set per ADOM, not per appliance

One FortiManager can hold normal-mode and backup-mode ADOMs side by side. That is the usual shape of a phased rollout: sites move into a backup-mode ADOM first so their configurations are captured, then into a normal-mode ADOM once the team is ready to manage them centrally.

04. ADOM Device Mode

A separate setting with a confusingly similar name. Operation mode governs *what FortiManager may do*; device mode governs *how finely a device can be split across ADOMs*.

Mode	Behaviour	When it applies
Normal	A FortiGate and all of its VDOMs belong to one ADOM together.	The default, and correct for most deployments.
Advanced	Individual VDOMs from one FortiGate can be assigned to different ADOMs.	Multi-tenancy on shared hardware — one physical firewall whose VDOMs belong to separate customers or business units.

```
config system global
    set adom-mode advanced
end
```

Advanced mode is effectively one-way

Returning to normal mode requires every VDOM assignment to be undone first, which on a populated appliance means unwinding the tenancy model itself. **Enable advanced mode only when the deployment genuinely requires per-VDOM separation**, not speculatively.

05. Creating an ADOM

The screenshot shows the 'Create ADOM' dialog box. The 'Name' field is 'Test-ADOM'. The 'Type' is 'FortiGate' with version '7.2' selected. The 'Description' field is empty. The 'Devices' section shows a table with columns 'Name', 'IP Address', and 'Platform', and a message 'No record found.' The 'Mode' section has 'Normal' selected. The 'General Management' section has 'VPN', 'FortiAP', and 'FortiSwitch' checked. The 'Default Device Selection for Install' section has 'Select All' and 'Deselect All' buttons. The 'Perform Policy Check Before Every Install' and 'Auto-Push Policy Packages When Device Back Online' sections have toggle switches. The 'Advanced Options' section is expanded, showing 'Time Zone' set to 'System Time Zone' and 'DNS' set to 'Use System DNS'.

Figure 11 — The Create ADOM dialogue.

Field	Note
Type and version	The device family and the FortiOS version the ADOM is pinned to — 7.2, 7.4 or 7.6 here. A device can only join an ADOM matching its firmware.
Devices	Devices can be assigned at creation or later. Empty here, since none are registered yet.
Mode	Normal or Backup, as described in section 3.
Central Management	Which managers are active inside this ADOM — VPN, FortiAP, FortiSwitch. Disabling one removes its pane from the ADOM entirely.
Perform Policy Check Before Every Install	Validates the policy package before installing. Off by default; worth enabling in production, where an install affects live firewalls.
Auto-Push Policy Packages When Device Back Online	Installs queued changes automatically when a device reconnects. Convenient for sites on unreliable links, and a surprise if nobody expects it.
Time Zone and DNS	Can be overridden per ADOM rather than inherited from the appliance — useful when ADOMs are split by geography.

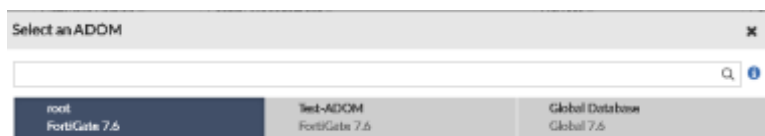


Figure 12 — The ADOM selector as the super-admin account: root, Test-ADOM and Global Database are all reachable.

06. Scoping an Administrator to an ADOM

An admin profile defines *what* an administrator may do. The Administrative Domain setting defines *where* they may do it. Both are needed: a read-only profile with access to all ADOMs still sees the entire estate.

Figure 13 — Creating an administrator scoped to a single ADOM, reusing the Lab-user profile from Lab 01.

Setting	Effect
All ADOMs	The account sees every ADOM on the appliance. Appropriate only for appliance-wide administrators.
All ADOMs except specified ones	An exclusion list. Useful where a team manages most of the estate but one tenant must stay private.
Specify	An inclusion list — the account sees only the named ADOMs. The multi-tenant default.
Admin Profile: Single / Per-ADOM	Per-ADOM assigns a different profile in each ADOM. One account can be read-write in its own ADOM and read-only in another — the normal shape for a site engineer who also needs visibility of a shared services ADOM.

Logging in as that account is the verification that matters.



Figure 14 — The scoped session. The banner reads ADOM: Test-ADOM and no ADOM selector is offered.

The account was taken directly into its assigned ADOM with no choice presented. Set against Figure 12, where the super-admin account is offered three ADOMs, this is a controlled demonstration that the scoping applied — not merely that it was saved.

07. ADOM Design and Versions

► 7.1 How to group devices

Grouping	Fits when
Firmware version	The recommended default. Every device in the ADOM accepts the same configuration syntax, so one policy package installs cleanly everywhere.
Region or site	Geographically distributed estates where local teams manage local firewalls.
Customer or tenant	MSSP deployments. Each customer's devices, policies and administrators are isolated from every other customer's.
Business function	Retail chains separating stores from distribution centres from head office, where each class of site has a different policy shape.

Firmware grouping is recommended for a concrete reason rather than a stylistic one: CLI syntax changes between FortiOS versions, so a policy package built for one version may not install cleanly onto a device running another. Grouping by version removes that class of failure entirely.

For an industrial estate the natural boundary is usually neither geography nor firmware but the architecture itself — an ADOM per plant, or per Purdue level, so that the firewalls enforcing a segmentation boundary are managed by the people accountable for it.

► 7.2 ADOM versions and upgrade

An ADOM's version is determined by the devices it holds, and the upgrade path is ordered.

- Upgrade the devices inside the ADOM first, then the ADOM itself — not the reverse.
- Upgrades proceed version by version. Skipping a major release is not supported.
- Migrating a device between ADOMs to achieve a version change is not the recommended path. The supported route is upgrading the device in place and then upgrading its ADOM.

This appliance holds no registered devices and is not being upgraded, so the mechanism is documented here rather than demonstrated. The constraint it creates is worth recording regardless: an estate running mixed firmware needs one ADOM per version, and the ADOM count is licensed.

o8. Health Check and Deletion

► 8.1 ADOM health check

A per-ADOM check reporting whether its devices are reachable and whether their configuration and policy state match what FortiManager holds.



Figure 15 — ADOM Health Check criteria.

Criterion	Reports
Device connection is down	The FGFM tunnel is not established. Always checked and cannot be disabled.
Device config status is not synchronized	FortiManager's device database and the device's running configuration disagree.
Device policy package status is not synchronized	The installed policy package does not match what the ADOM holds.
Exclude model devices when check	Suppresses pre-staged devices that are not yet online, which would otherwise report as failures by design.

The second and third criteria are the states covered in Lab 05. The health check is the fastest way to find them across a whole ADOM rather than device by device.

► 8.2 Why an ADOM would not delete

26 Task failed with error: Unable to delete because the adom is in global assignment list or is referenced by a defined admin account

Figure 16 — The deletion task failing in Task Monitor, with the reason.

Symptom	Deleting the ADOM failed. The task completed in two seconds with an error rather than hanging or partially applying.
Cause	The error names it: the ADOM is in the global assignment list, or it is referenced by a defined admin account. The Test-Adom administrator created in section 6 is scoped to this ADOM, and that reference holds it.
Fix	Remove or re-scope every administrator referencing the ADOM, and clear any global database assignment, then delete.
Rule	FortiManager will not delete an object another object depends on. The Task Monitor entry names the dependency, which makes it the first place to look rather than the last — the distinction drawn in section 1.5.

09. Summary

Area	Outcome	Evidence
Architecture and protocols	Documented	FGFM on TCP 541 for management; FortiGuard protocol for update distribution
Version compatibility	Verified	Compatibility Tool confirming FortiManager 7.6.6 against FortiOS 7.0
ADOMs enabled	Enabled	ADOM selector present after re-login
Default ADOMs	Documented	root, Global Database, and pre-created ADOMs per device family
Licence consumption	Finding	0/2 with all defaults present; 1/2 after one custom ADOM — only custom ADOMs count
Operation mode	Normal and Backup documented	Mode selector; behaviour difference recorded
Device mode	Normal and Advanced documented	VDOM-per-ADOM scenario recorded
ADOM created	Created	Test-ADOM, FortiGate 7.6, normal mode
Administrator scoping	Verified by login	Scoped session opening directly into its ADOM with no selector offered
ADOM deletion	Finding	Blocked by an administrator account referencing the ADOM; named in Task Monitor
Health check	Documented	Criteria recorded; device-dependent checks deferred to Lab 05